OPERATORS: **Operator is a special symbol designed for a particular task [work]. C comes with 44 operators and 14 separators. Operator works on operands. Based on no of operands participating in operation, the operators divided into 3 types.**

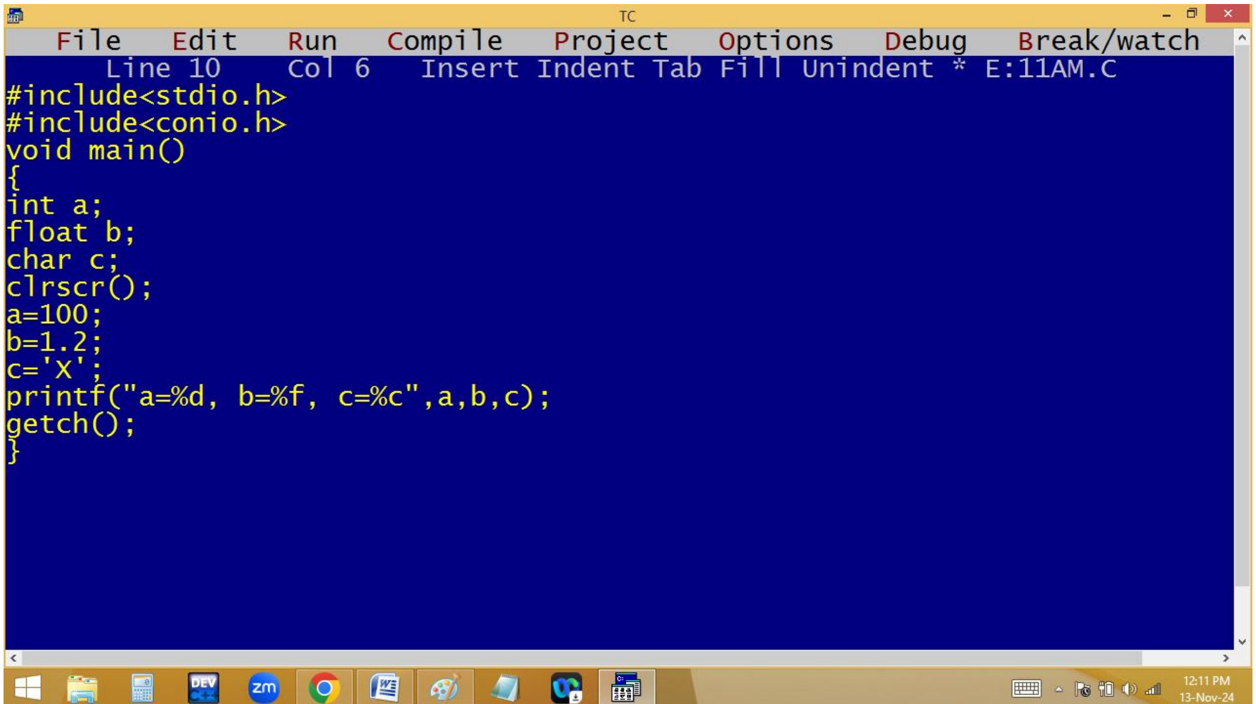
1. **Unary operator: Require one operand.**
   Eg: a++, a--, ++a, --a, sizeof(a), ~a, !a, +a, -a,…
2. **Binary operator: Require two operands.**
   Eg: a+b, a>b, a<=b, a!=b, a<<b,….
3. **Ternary / conditional operator [? :] : Require 3 operands.**
   Eg: conditional part ? true part : false part ;

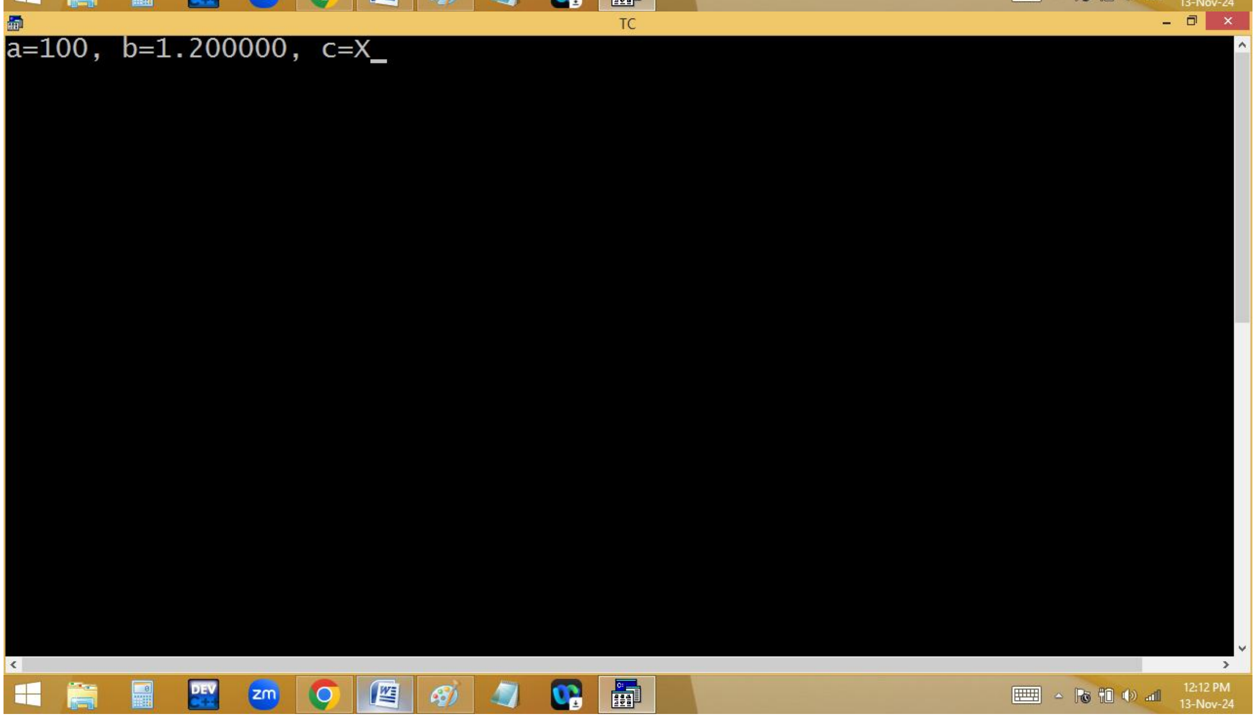
   5 > 9 ? ” 5 is big “ : “ 9 is big “;

   7 > 2 ? “7 is big” : “ 2 is big “ ;

Based on operation, the operators divided into several types.

1. **Assignment operator [=]: It copies the value on its right side into the variable on its left side. In assignment operation, the left side operand should be a variable.** i.e. expressions and constants not allowed on left side.
   Eg:
   a=100;
   b=1.2;
   c=’x’;
   d=”abc”; ➔ Error ➔ string copy not allowed with =


TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 6 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
float b;
char c;
clrscr();
a=100;
b=1.2;
c='X';
printf("a=%d, b=%f, c=%c",a,b,c);
getch();
}
12:11 PM
13-Nov-24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a=100, b=1.200000, c=X_
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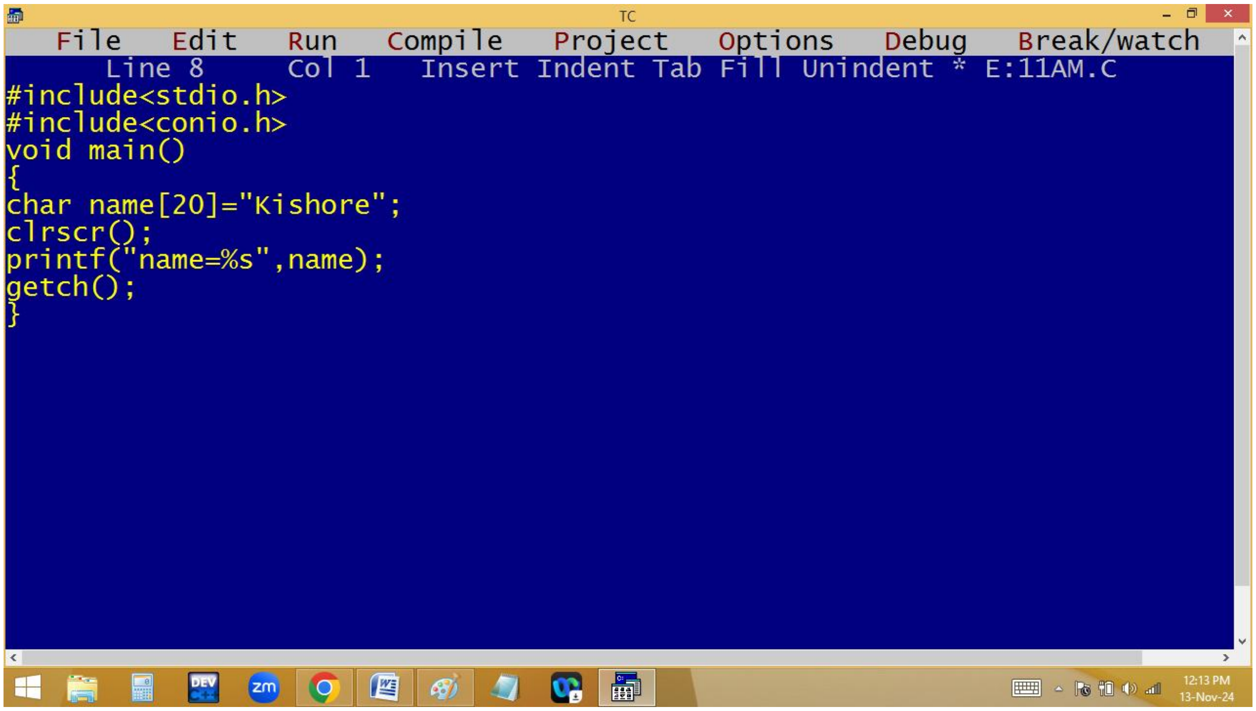
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#include<stdio.h>
#include<conio.h>
void main()
{
char name[20]="Kishore";
clrscr();
printf("name=%s",name);
getch();
}
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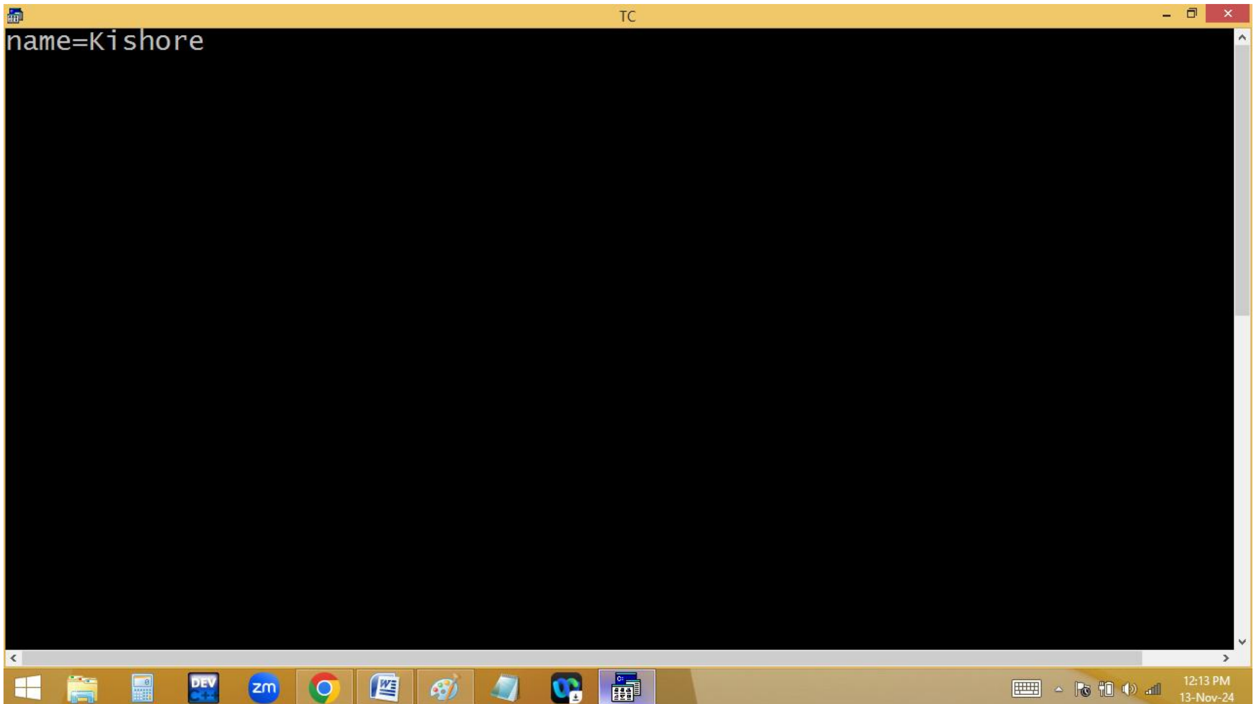
13-Nov-24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name=Kishore
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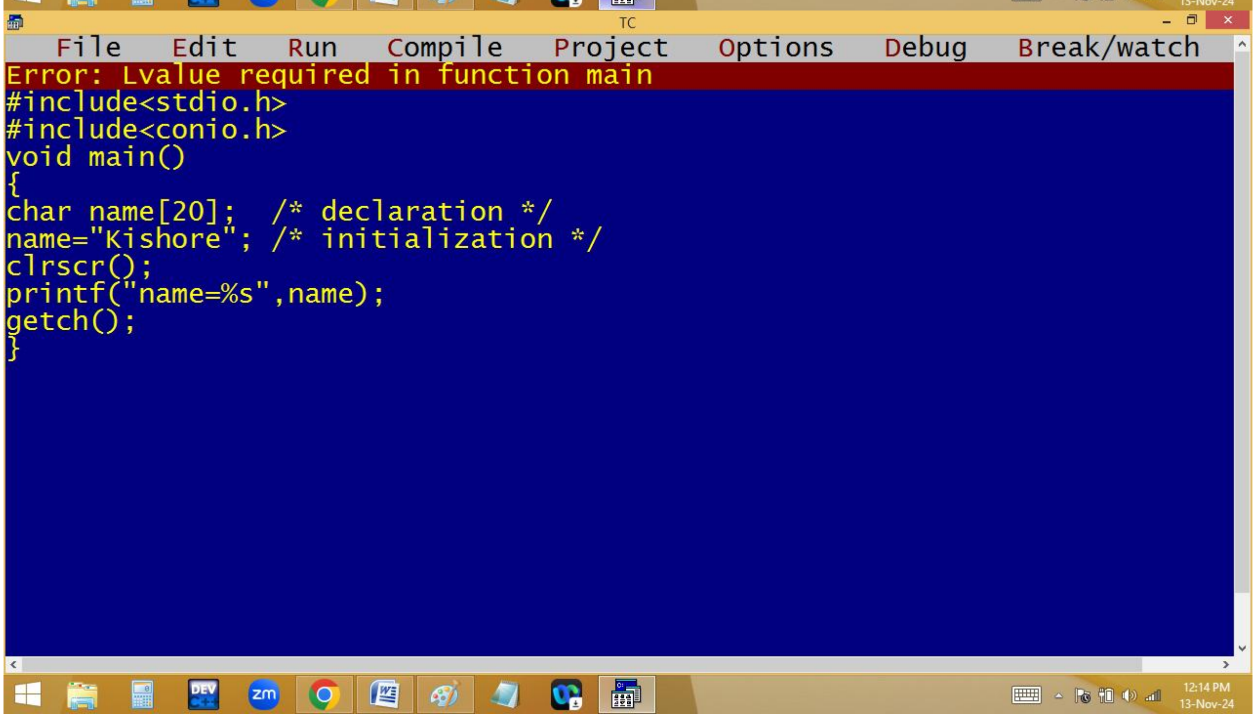
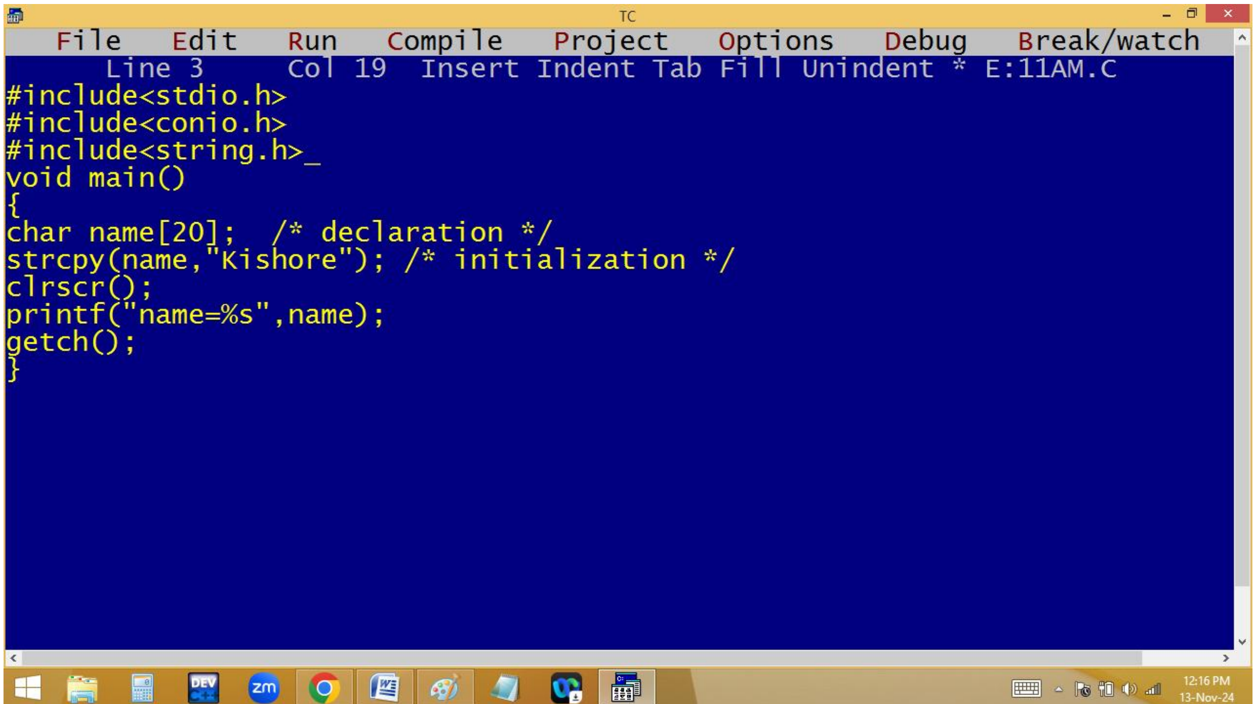

TC
File Edit Run Compile Project Options Debug Break/watch
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
char name[20]; /* declaration */
name="Kishore"; /* initialization */
clrscr();
printf("name=%s",name);
getch();
}
12:14 PM
13-Nov-24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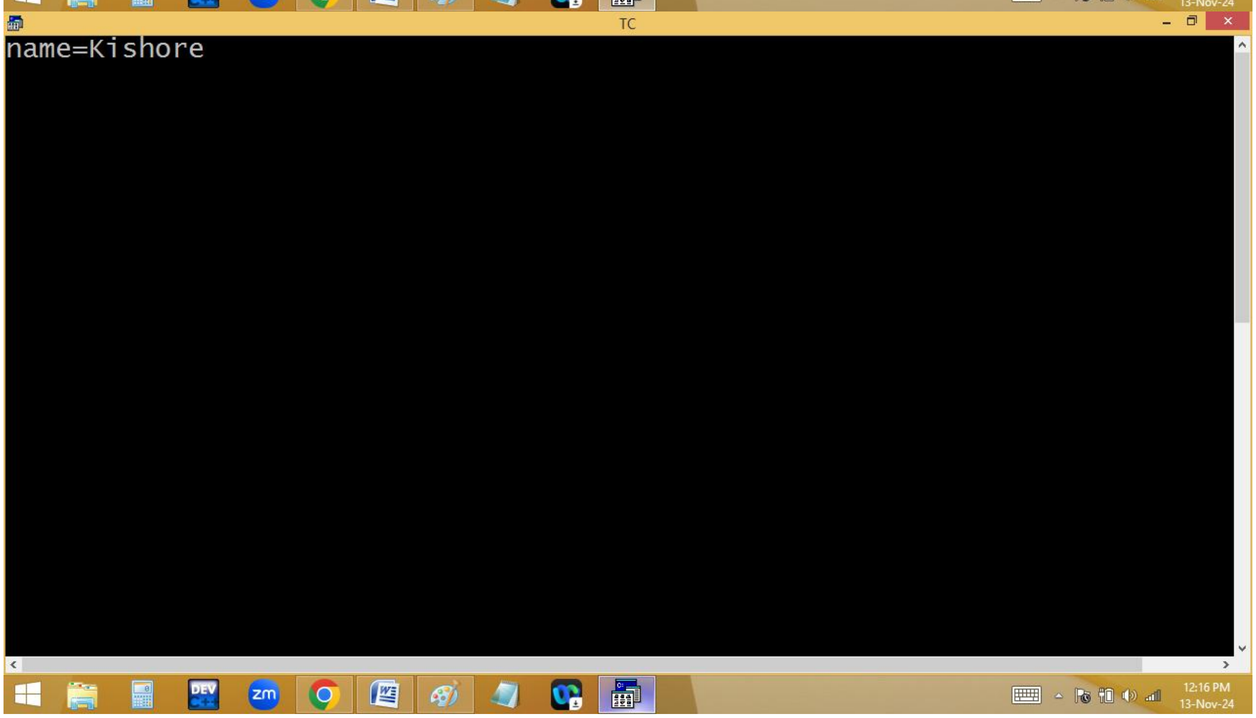
Line 3 Col 19 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
#include<string.h>_
void main()
{
char name[20]; /* declaration */
strcpy(name,"Kishore"); /* initialization */
clrscr();
printf("name=%s",name);
getch();
}
12:16 PM
13-Nov-24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name=Kishore
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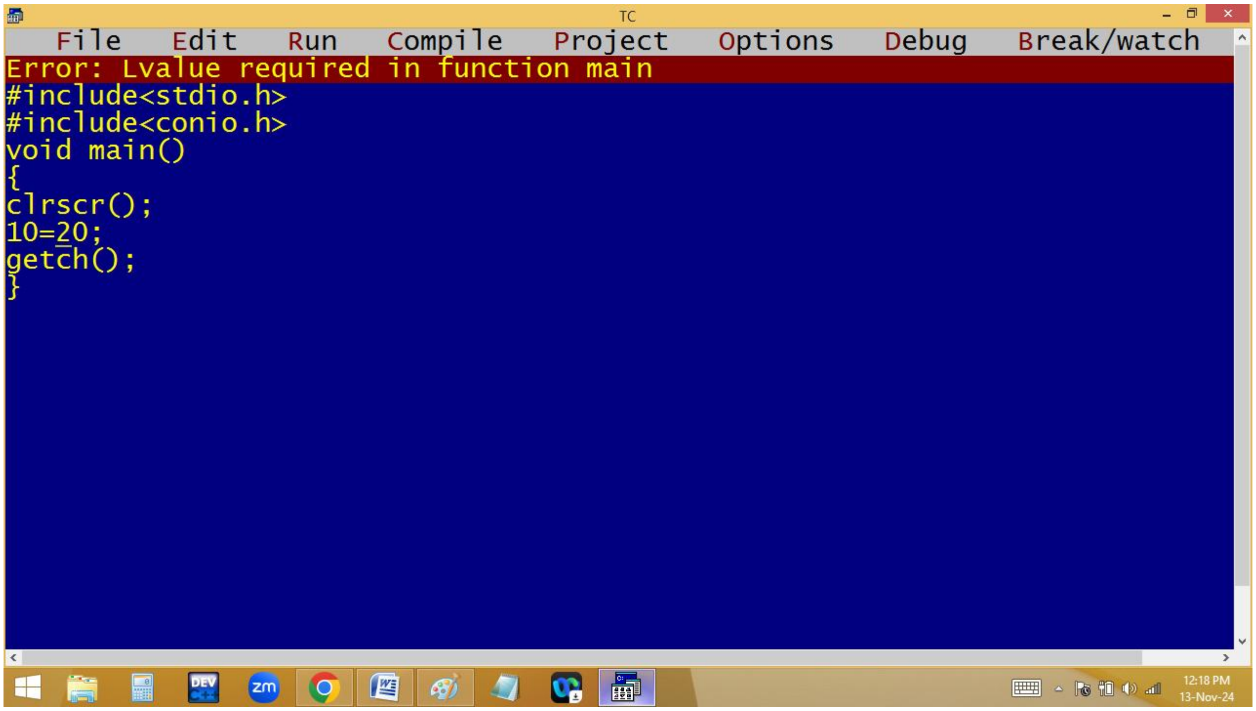
10=20; → Error


TC
File Edit Run Compile Project Options Debug Break/watch
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
10=20;
getch();
}
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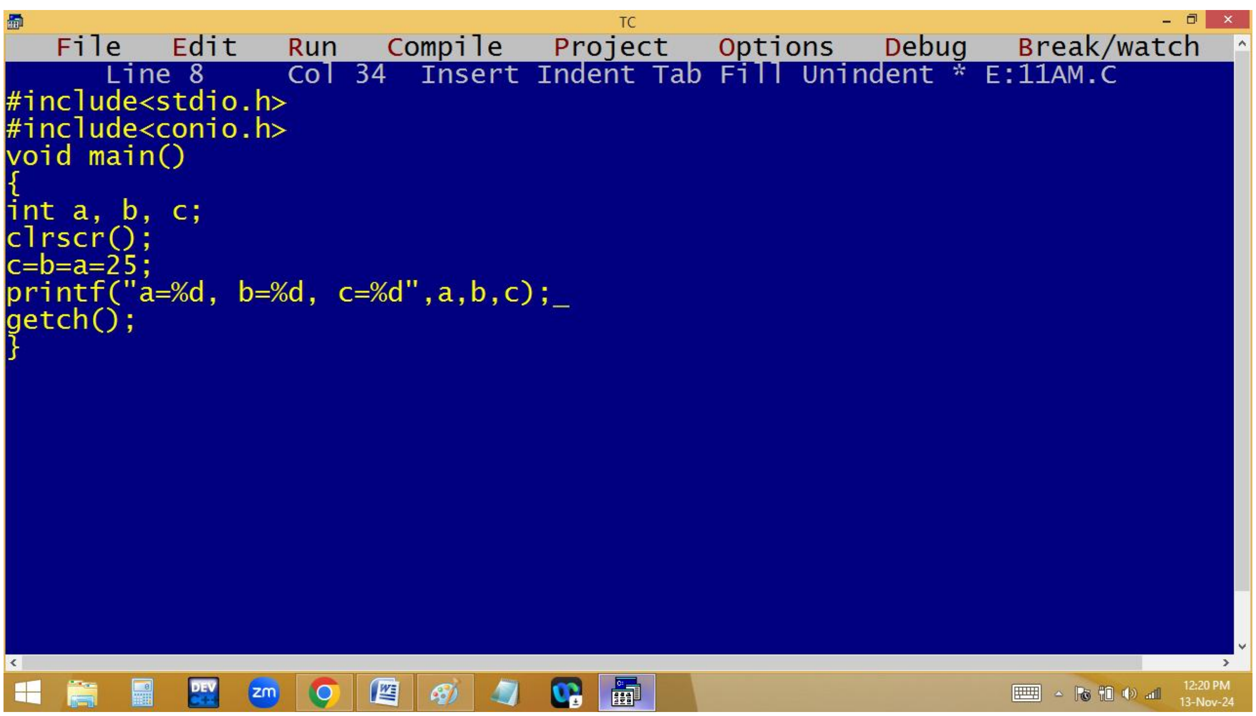
13-Nov-24

a=b=c=25;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Line 8 Col 34 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b, c;
clrscr();
c=b=a=25;
printf("a=%d, b=%d, c=%d",a,b,c);_
getch();
}
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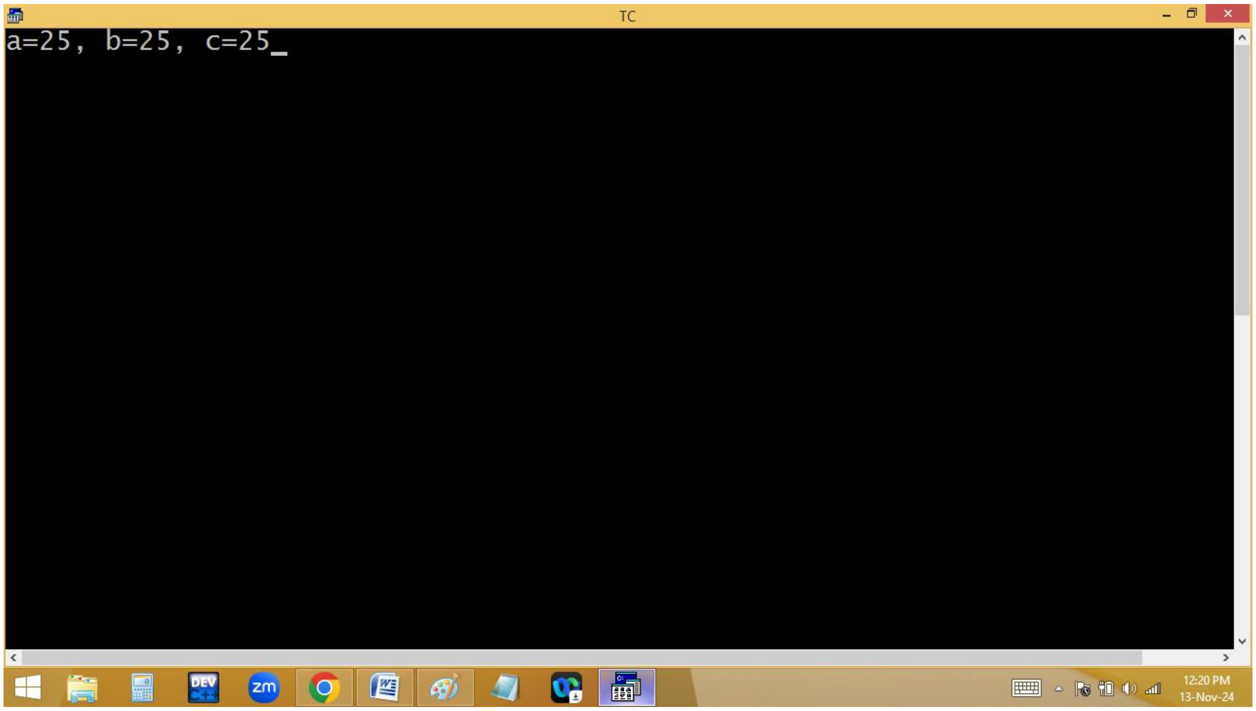
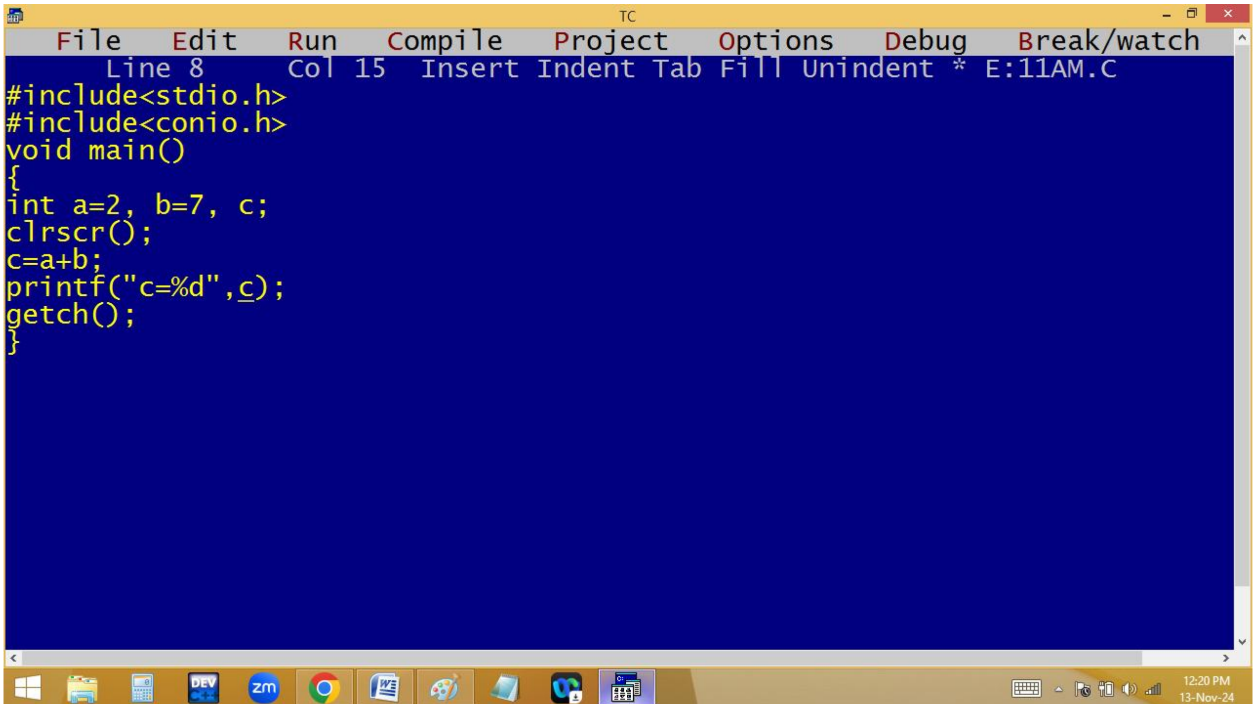
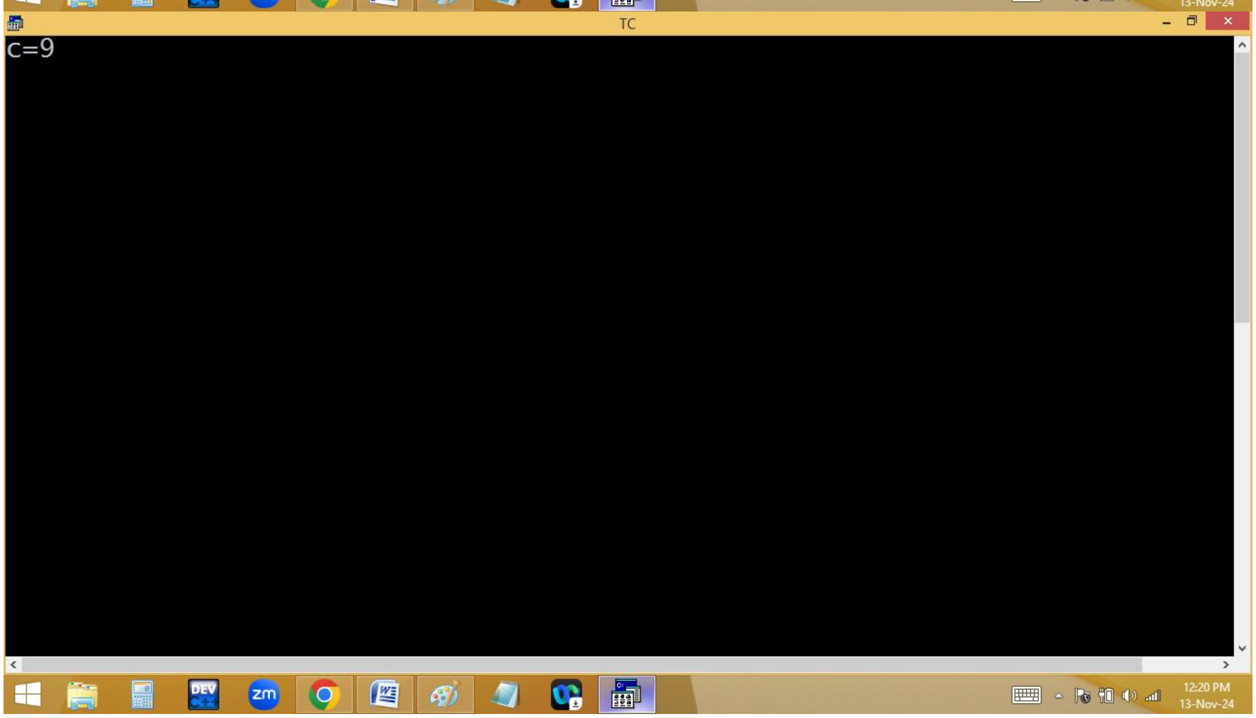


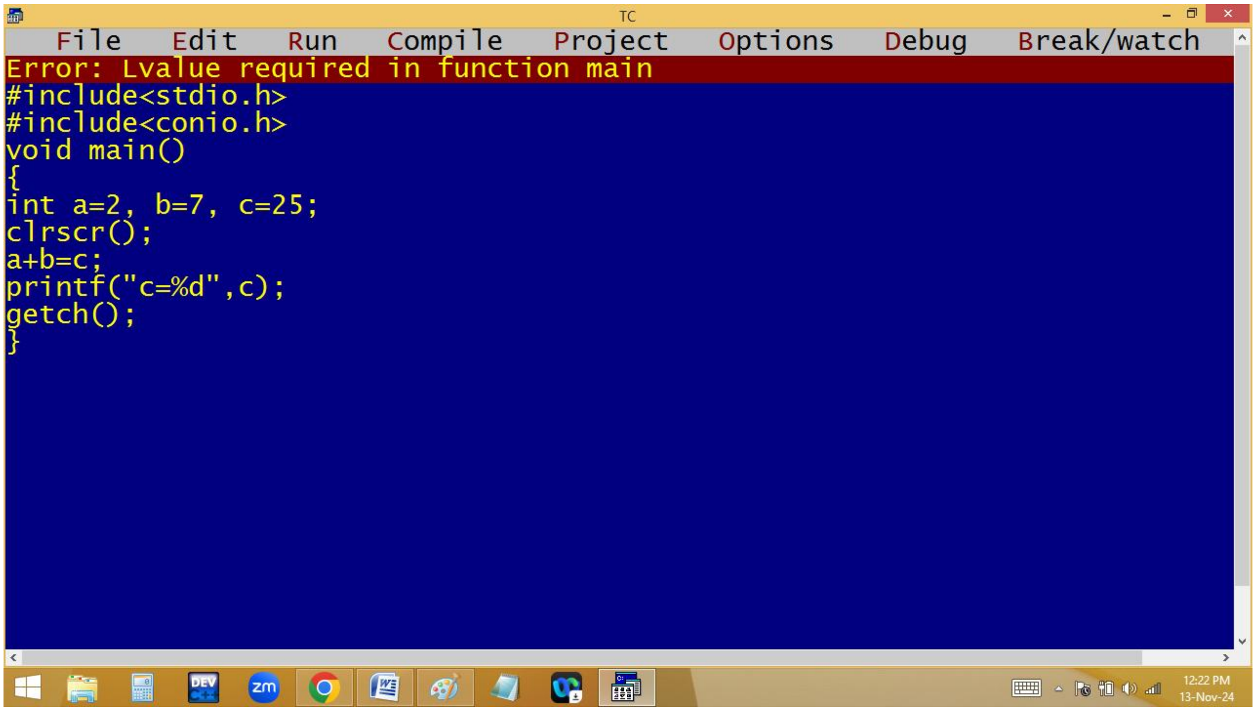
c=2+7 ; → c=9

2+7= 9; → Error



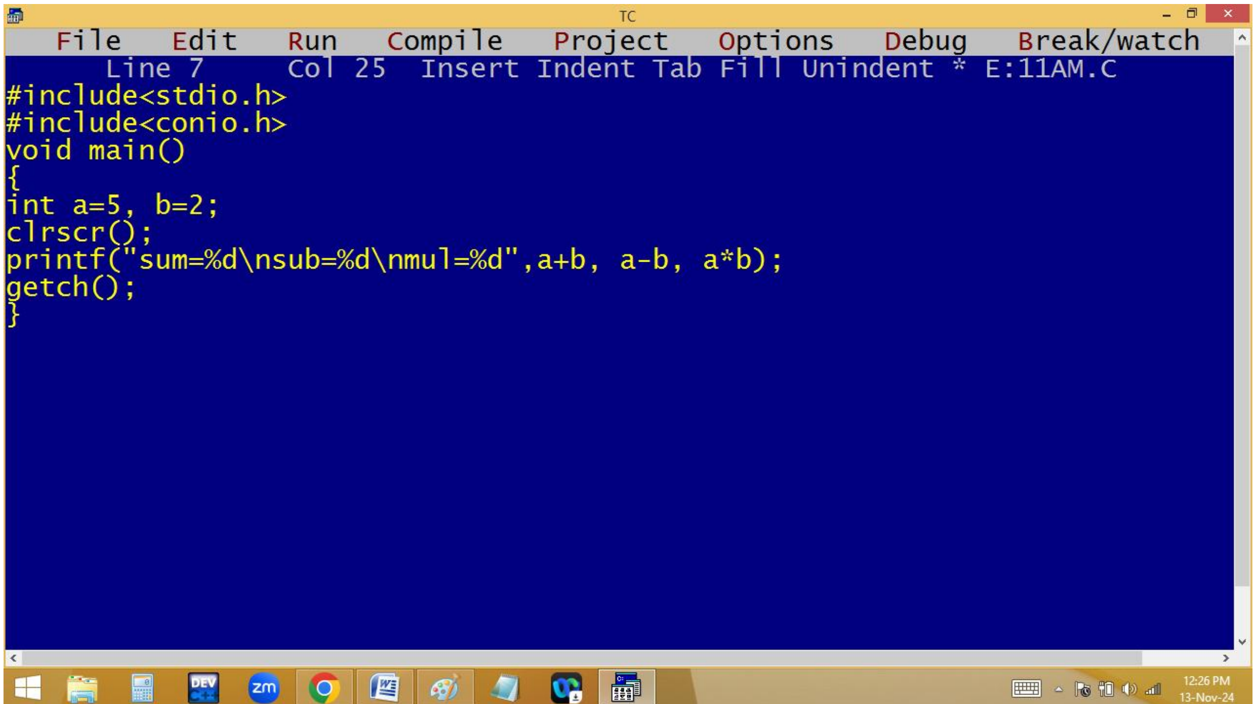

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a=2, b=7, c;
clrscr();
c=a+b;
printf("c=%d",c);
getch();
}
```

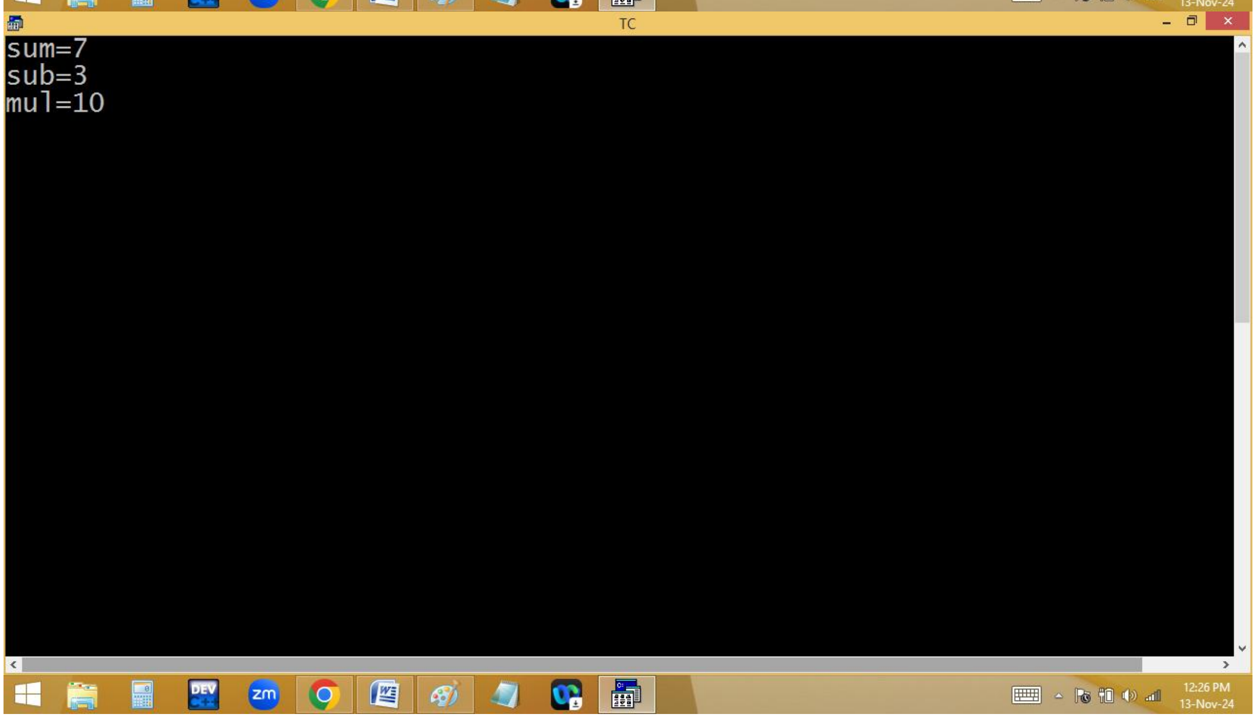




2. **Arithmetic operators [+, -, *, %, /]: They are used for mathematical operations.**

Eg: a+b, a-b, a*b,….

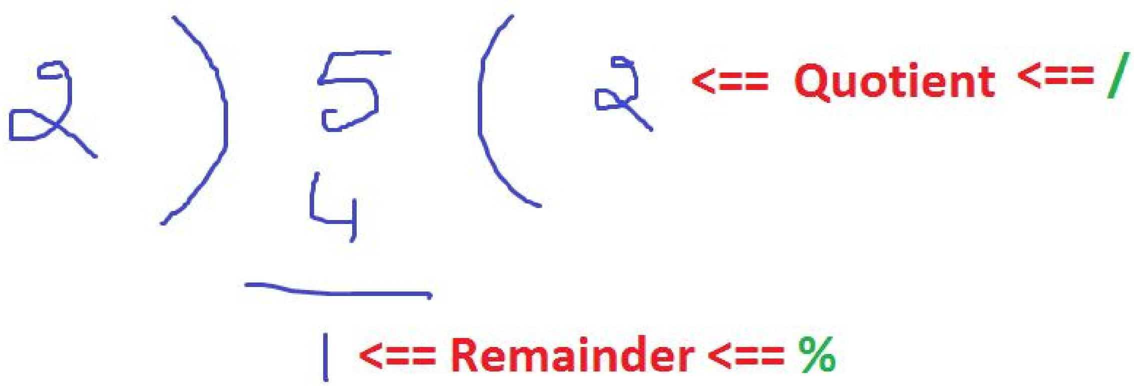




% - modules [Remainder]

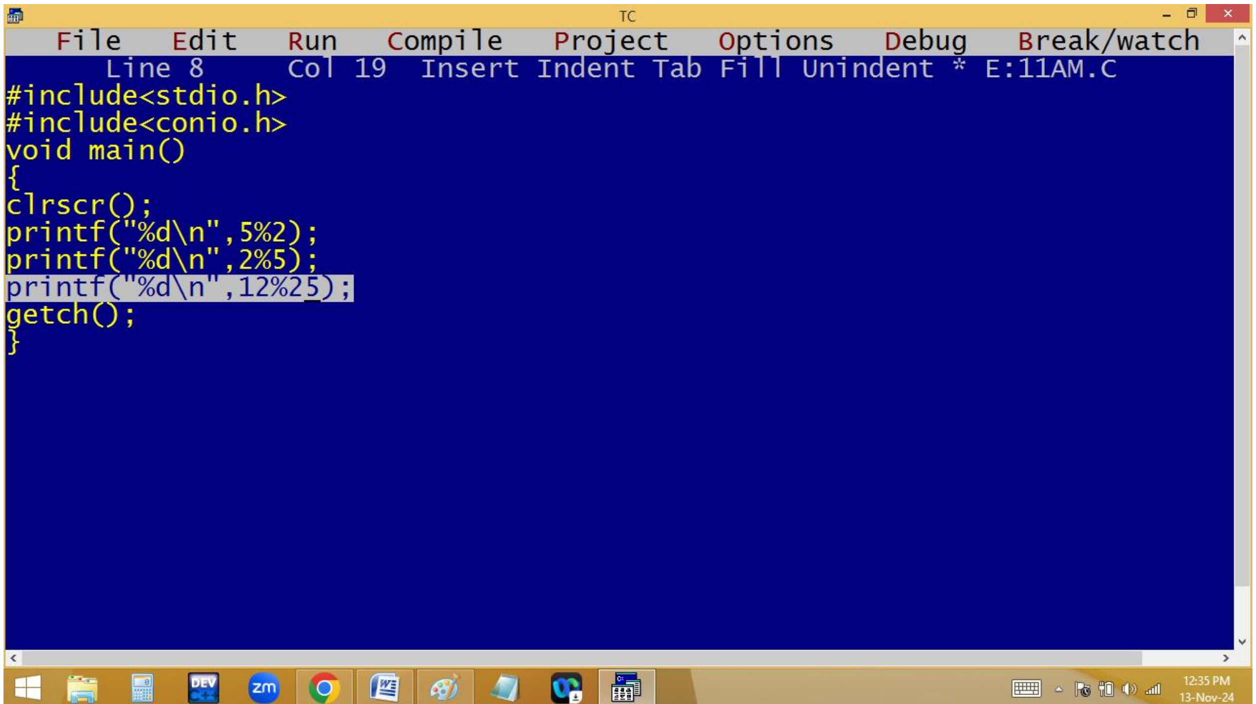
Eg:

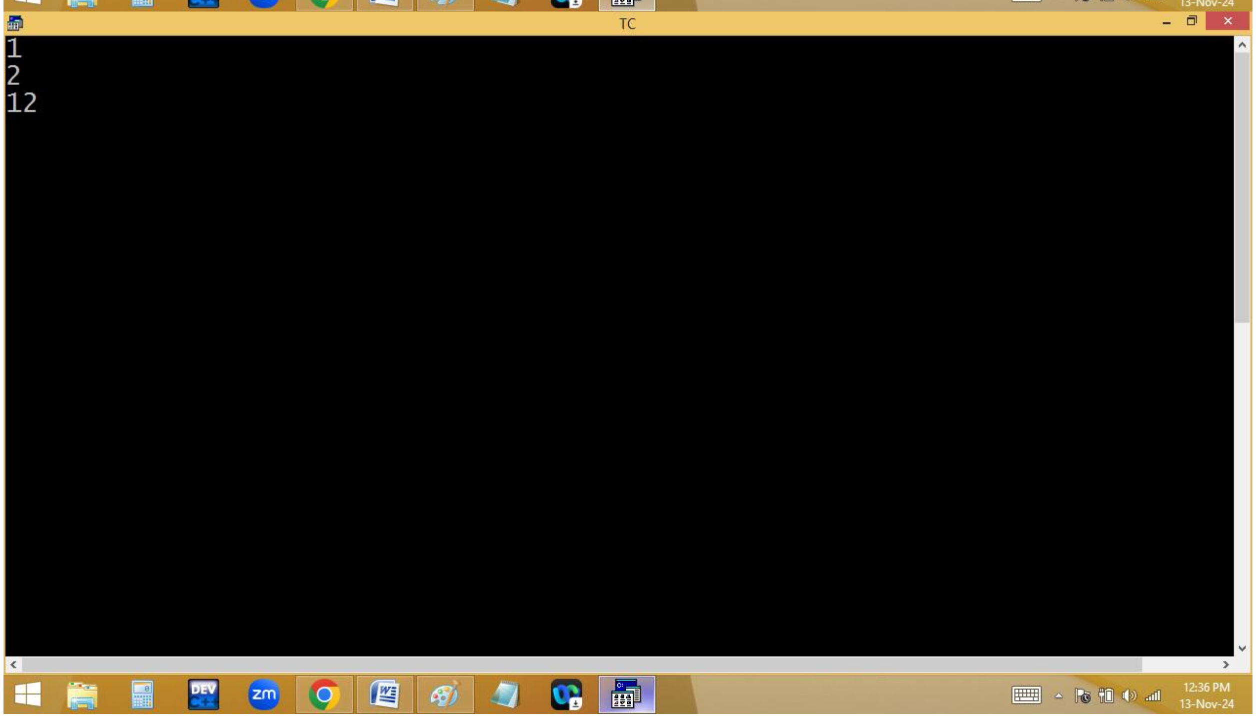
5 % 2 = 1



2 % 5 = 2

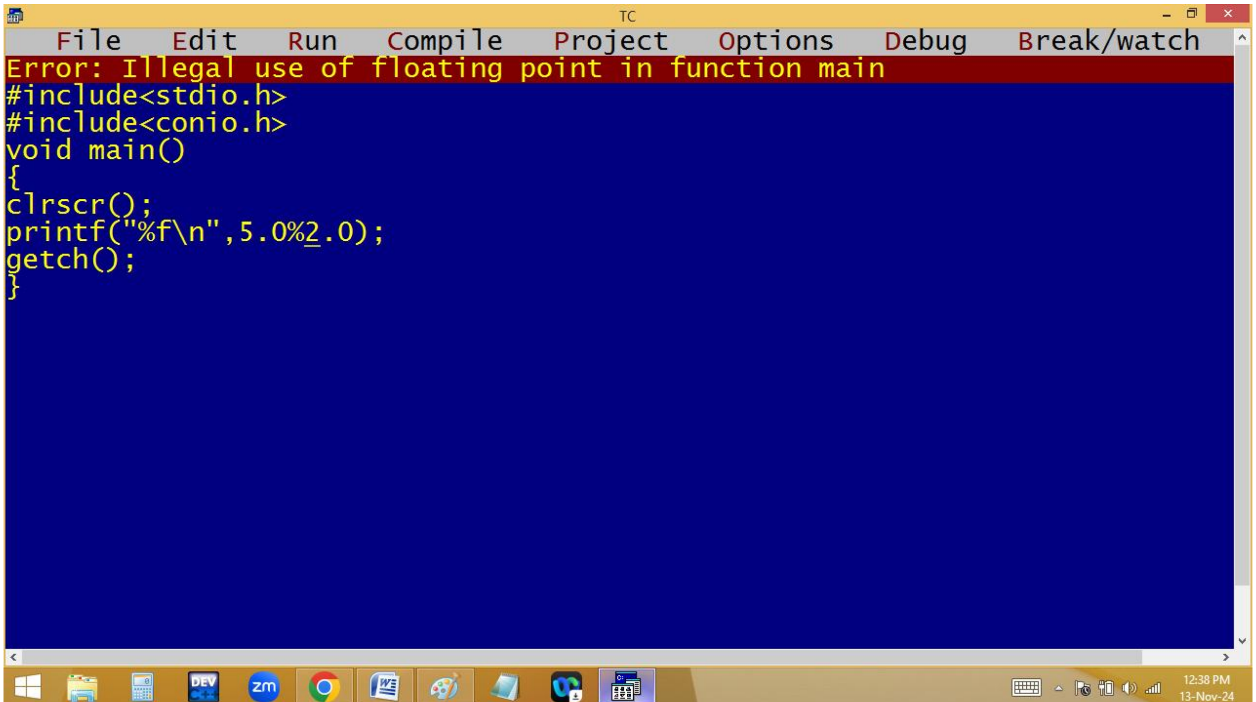
Note: If the divisor is bigger than dividend then the dividend is the answer.





5.0 % 2.0 = Error

Note: We can’t do floating modules with % operator. For this we have to use fmod() available in <math.h>


TC
File Edit Run Compile Project Options Debug Break/watch
Error: Illegal use of floating point in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f\n",5.0%2.0);
getch();
}
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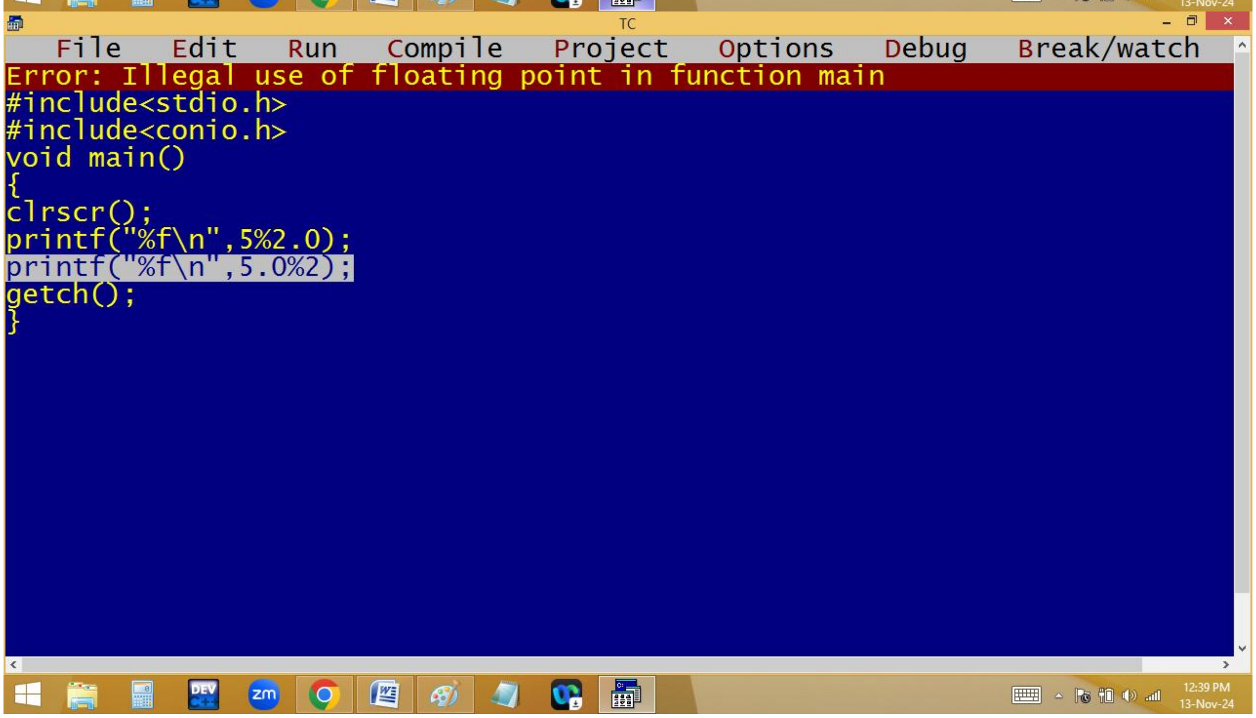
13-Nov-24


TC
File Edit Run Compile Project Options Debug Break/watch
Error: Illegal use of floating point in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f\n",5%2.0);
printf("%f\n",5.0%2);
getch();
}
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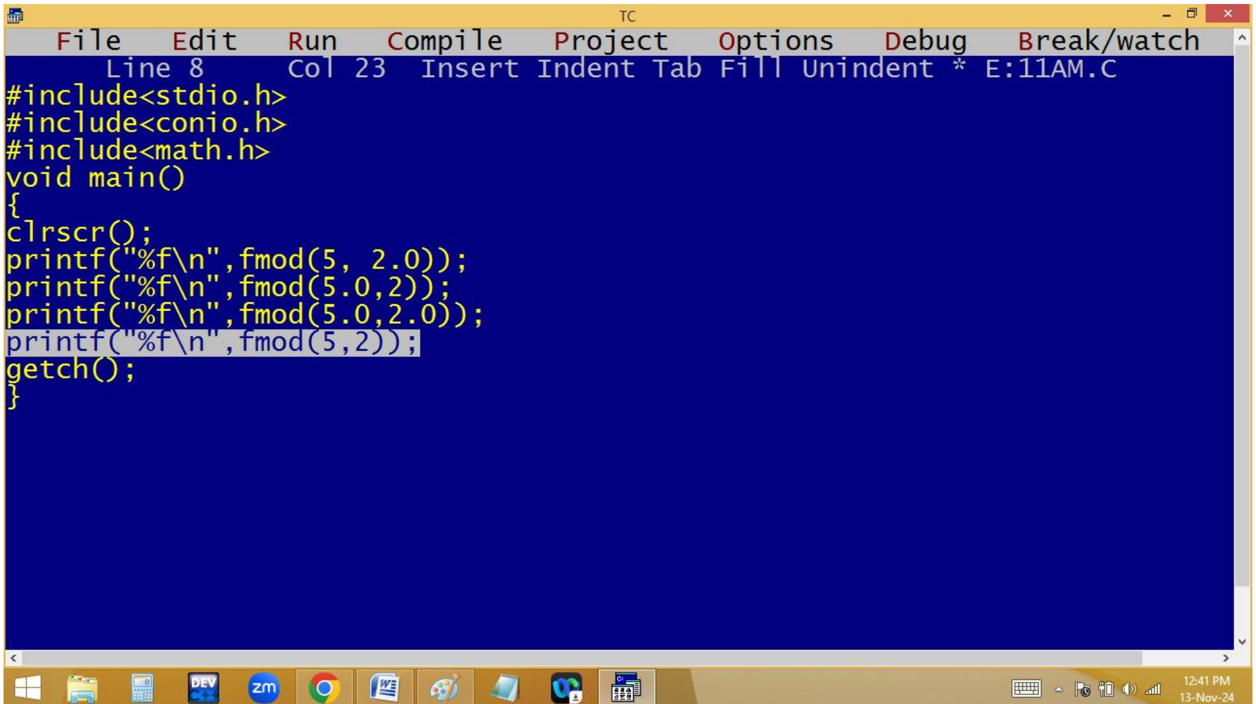
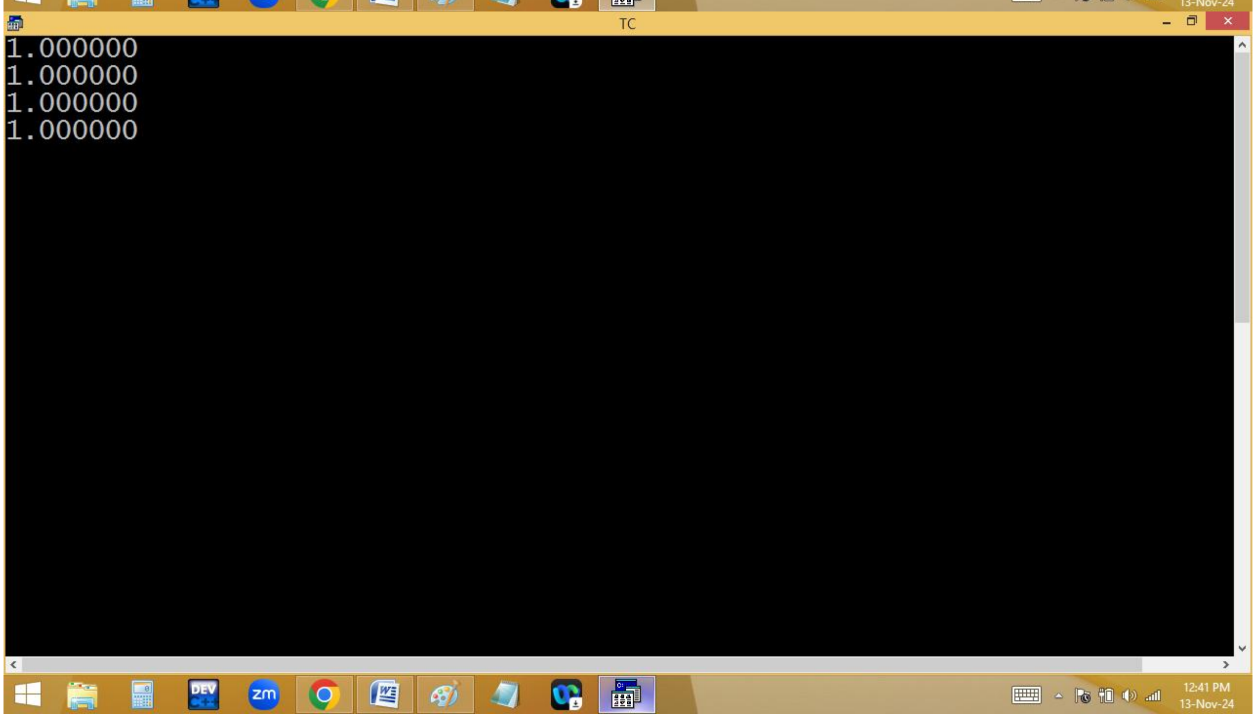
13-Nov-24



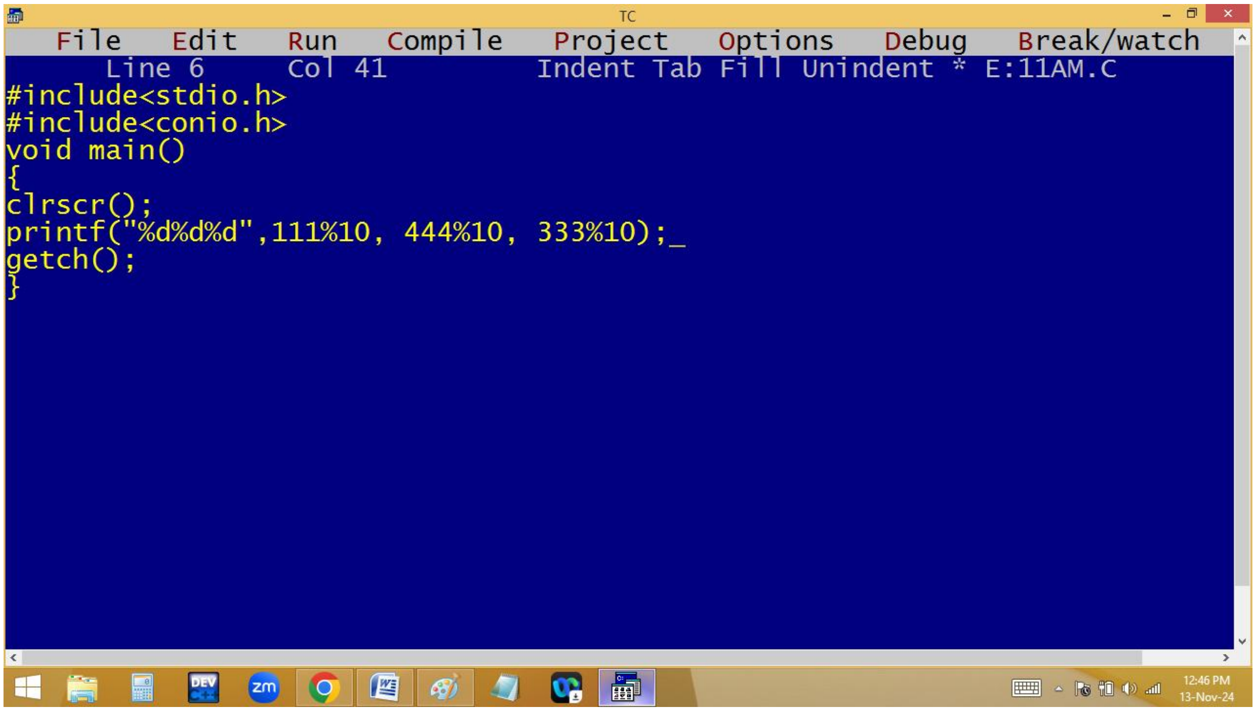


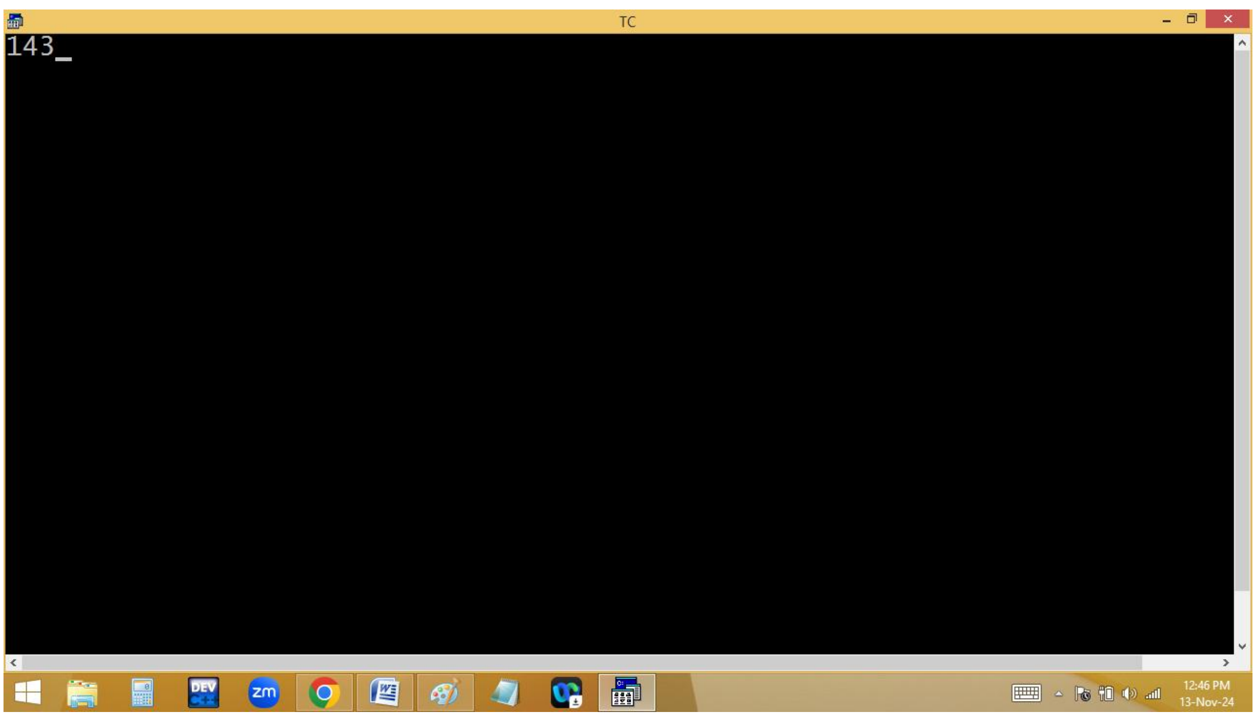
111%10=1

444%10=4

333%10=3

Note: Any no%10 gives last digit.





-5%2= -1

5%-2= 1

-5%-2= -1

Note: if the numerator is negative then result also negative.